



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

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SECP3213 BUSINESS INTELLIGENCE

SECTION 01

PROJECT PHASE 1 INTRODUCTION

Analysing the Impact of Macroeconomic Indicators and Financial

Performance on Malaysian Banking Stocks

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1. Problem Background

1.1 Business context

The banking sector has always been one of the foundational pillars of the economy of Malaysia. Popular banking service providers such as Maybank, CIMB Bank and RHB Bank serve as a key indicator of a broader economic health. For investors and financial analysts, evaluating banking stocks requires looking beyond just daily price movements. A bank's market valuation is intricately tied to its internal financial health such as Revenue, Net Profit, and ROE, heavily influenced by domestic and global macroeconomic factors, including Malaysia's Gross Domestic Product (GDP) the Overnight Policy Rate (OPR) and the U.S. Federal Funds Effective Rate (DFF).

1.2 Importance of the problem

The data required to perform a comprehensive analysis of banking stocks is highly distributed and siloed. Financial performance data, daily trading volumes and macroeconomic indicators are scattered across multiple platforms and separated by multiple banks. Analyzing these datasets in isolation often leads to an incomplete understanding of market dynamics. For example, a bank may report strong earnings, but external macroeconomic headwinds like an interest rate hike could still trigger a stock sell-off. Without a centralized, integrated view of how these distinct data points interact, stakeholders struggle to accurately assess volatility, identify correlations and forecast future performance.

1.3 Expected impact

By integrating these disparate datasets into a unified Business Intelligence (BI) solution, this project will bridge the gap between micro-level financial data and macro-level economic indicators. The resulting dashboards will allow users to visualize trends, compare the financial resilience of the selected three Malaysian banks and uncover how specific economic shifts impact stock prices. Ultimately, this analysis shall facilitate highly informed, data-driven investment strategies and improved risk management.

2. Objectives

This project aims to achieve the following objectives:

1. To analyse and compare the fundamental financial performance (Revenue, Net Profit, EPS, ROE) and historical stock price trends of Maybank, CIMB, and RHB over time.
2. To determine the correlation and impact of macroeconomic indicators (Malaysia's OPR, GDP, and the U.S. Federal Funds Rate) on the stock price volatility and trading volumes of Malaysian banks.
3. To design and develop an interactive Business Intelligence dashboard that combines structured time-series, financial and macroeconomic data into a unified platform for seamless data exploration and trend visualization.
4. To perform time-series analysis and forecasting on banking stock prices by leveraging historical financial trends and economic data combinations.

3. Dataset Description

3.1 Dataset 1: Daily Stock Price Dataset

This dataset consists of historical daily stock price data for Malayan Banking Berhad, CIMB Group Holdings Berhad, and RHB Bank Berhad.

1. Source of dataset

This dataset is obtained from [Investing.com](https://www.investing.com), which provides historical stock market information for publicly listed companies.

2. Type of dataset

The dataset is classified as structured time series data because it contains daily observations recorded over a period of time.

3. Key Variables

Variables	Description	Data Type
Date	Trading date	Date
Open	Opening stock price	Float
Price	Closing stock price	Float
Volume	Number of shares traded	Float
High	Highest stock price of the day	Float
Low	Lowest stock price of the day	Float
Stock Name	Stock ticker	String

4. Sample Data

Raw data

All three banks have the same data variables.

Date	Price	Open	High	Low	Vol.	Change %
5/7/2026	11.26	11.24	11.36	11.22	28.73M	0.36%

Processed data

Date	Open	Close	Vol. (M)	High	Low	Stock Name
5/7/2026	11.24	11.26	28.73	11.36	11.22	MAYBANK
5/7/2026	7.96	8.04	37.96	8.15	7.96	CIMB
5/7/2026	8.31	8.33	9.08	8.40	8.29	RHBBANK

3.2 Dataset 2: Financial Performance Dataset

This dataset focuses on financial performance of Malayan Banking Berhad, CIMB Group Holdings Berhad, and RHB Bank Berhad.

1. Source of dataset

This dataset is obtained from [KLSE Screener](#), which provides financial data and market-related information for companies listed on Bursa Malaysia.

2. Type of dataset

The dataset is classified as structured financial data as it contains organized numerical information related to company performance.

3. Key Variables

Variables	Description	Data Type
Stock	Stock ticker	String
Year	Financial reporting year	Date
Q Date	Specific quarter-end reporting date	Date
Quarter	Financial reporting quarter	Integer
Announced	Date when the financial results were officially announced to the public	Date
EPS	Earnings Per Share (EPS). Profit allocated per share	Float
DPS	Dividend Per Share (DPS). Representing dividends distributed per share.	Float
NTA	Net Tangible Assets per share (NTA). Showing the value of physical assets after liabilities	Float
Revenue (M)	Total income generated by the bank during the quarter (in millions)	Integer
P/L (M)	Profit or Loss earned during the quarter (in millions)	Float
ROE	Return on Equity(ROE). Measures profitability relative to shareholders' equity.	Float
QoQ%	Quarter-over-Quarter growth percentage (QoQ%). Showing performance changes compared to the previous quarter.	Float
YoY%	Year-over-Year growth percentage (YoY%). Showing performance changes compared to the previous year.	Float

4. Sample Data

Raw data

All three banks have the same data variables.

EPS	DPS	NTA	Revenue	P/L	Quarter	Q Date	Financial Year	Announced	ROE	QoQ %	YoY %
22.1500	33.000	7.7348	15.8b	2.7b	4	2025-12-31	31 Dec, 2025	2026-02-26	2.9%	2.1%	5.7%

Processed data

Stock	Financial Year	Q Date	Quarter	Announced	EPS	DPS	NTA	Revenue (M)	P/L (M)	ROE	QoQ %	YoY %
CIMB	2025-12-31	2025-12-31	4	2026-02-27	17.8	20.35	6.5204	5400	1900	0.027	0.076	0.066

3.3 Dataset 3: Economic Indicators Dataset

This dataset consists of macroeconomic indicators that may influence the performance of banking stocks.

1. Source of dataset

This dataset is obtained from multiple sources, including [Bank Negara Malaysia](#) for Overnight Policy Rate (OPR) information, [OpenDOSM](#) for Gross Domestic Product (GDP) growth data and [Federal Reserve Bank of St. Louis \(FRED\)](#) for Federal Funds Effective Rate (DFF).

2. Type of dataset

The dataset is classified as structured macroeconomic data because they contain organized economic indicators collected periodically.

3. Key Variables

Variables	Description	Data Type
Year	Economic record year	Date
OPR	Overnight Policy Rate(OPR). Malaysia's benchmark interest rate set by Bank Negara Malaysia.	Float
GDP	Quarterly Gross Domestic Product(GDP) growth rate obtained from Department of Statistics Malaysia via OpenDOSM	Float
DFF	Federal Funds Effective Rate(DFF). U.S. interest rate obtained from Federal Reserve Bank of St. Louis FRED database	Float

4. Sample Data

Raw data

OPR Decision and Statement from Bank Negara Malaysia

Date	Change in OPR (%)	New OPR Level (%)
06 Nov 2025	0	2.75

Quarterly Real GDP by Economic Sector from Department of Statistics Malaysia

series	date	sector	value
abs	1/1/2015	p0	282255

Lookup Table: GDP

method	code	desc_en	desc_bm
production	p0	GDP at purchasers' prices	KDNK pada harga pembeli

Federal Reserve Bank of St. Louis

observation_date	DFF
5/12/2021	0.06

Processed data

Year	OPR	GDP	DFF
2015	2.75	282255	0.06

4. Proposed BI Approach

4.1 Data Ingestion

All collected datasets will be imported into Alteryx using the Input Data tool.

4.2 Data Cleaning

The following data cleaning processes will be performed to improve data quality and consistency:

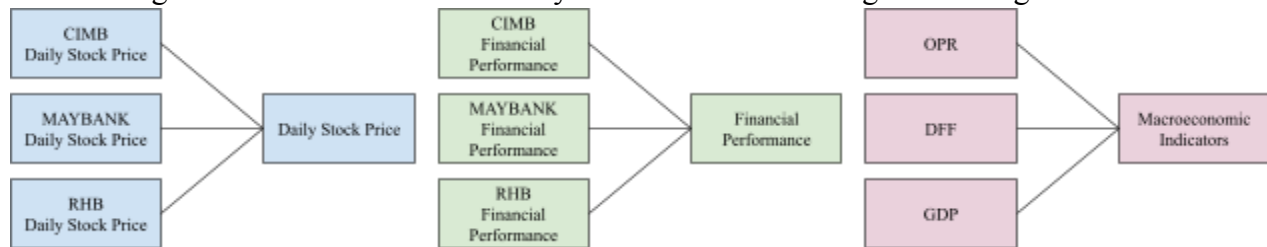
- Standardise date formats using the “DateTime” tool
- Standardise numeric values with suffix including ‘b’ and ‘m’, representing billion and million using “Formula” tool
- Handle null values depending on the need
- Remove rows with null values in the financial performance datasets using the “Filter” tool

- Apply forward fill for missing economic indicator values using the “Multi-Row Formula” tool

4.3 Data Integration

Data from multiple sources will be integrated into a unified dataset for analysis.

- Merge daily stock price datasets using the “Union” tool
- Combine financial performance datasets using the “Union” tool
- Integrate the economic indicators by date based on matching dates using the “Join” tool



4.4 Data Transformation

Data transformation processes will be conducted to prepare the datasets for analysis and visualisation.

- Create calculated fields such as
 - Daily stock return
 - Percentage price change
 - Moving averages
 - Revenue growth rate
 - Net profit margin
- Use “Formula” and “Summarise” for calculation and aggregation

4.5 Proposed Analysis

The analysis aims to identify relationships between stock price performance, company financial performance and macroeconomic indicators.

Analysis includes:

- Stock price trend analysis
- Comparison of financial performance between banks
- Correlation analysis between stock prices and economic indicators
- Impact of OPR and GDP growth on banking stock performance
- Volatility and trading volume analysis
- Time-series forecasting of stock prices based on historical trends

4.6 Proposed Visualisations

Visualisations are proposed as follows:

- Line chart to show stock price trend over time
- Candlestick chart to show daily stock price movement
- Bar chart to compare revenue, net profit and EPS among banks
- Scatter plot to analyse relationship between stock prices and economic indicators
- Heatmap to show correlation between variables such as OPR, GDP, stock return and trading volume